

Unit 405 - Field Data Review

Mohawk Valley West Slope demonstration. Method **Cable**, 42.6 ac, BAF 20 variable-radius cruise on a 300 ft grid, 20 plots, 236 trees tallied. Simulated data with planted recording errors; see docs/methods.md.

Table 1. Stand summary from the BAF 20 cruise, per acre

STAND DENSITY		STAND INFO	
Basal area	236 sq ft/ac (SE 2.3 %)	Type	Sierran mixed conifer
Trees per acre	61	CWHR size / density	5 / D (cover 95 %)
QMD	26.6 in	SDI	278 (47 % of max 593)
SAWTIMBER (>= 10 in)		BIOMASS (< 10 in)	
BA QMD	235 sq ft/ac 27.0 in	BA QMD	1 sq ft/ac 9.5 in
TPA	59	TPA	2
Volume	63.2 MBF/ac (11484 cu ft)	Volume	0.6 green tons/ac (21 cu ft)
TREATMENT TARGET		CRUISE DESIGN	
Leave	35 % of max SDI = ~176 sq ft/ac BA	Sampling error	4.7 % at 95 % (t = 2.09, n = 20)
Species (BA)	WF 75, PP 57, IC 38, DF 37, SP 29	Stratum standard	40 % - MEETS; 20 plots vs 20-plot practice: ok

Cubic volume by form factor 0.42 net of defect; Scribner board feet at 5.5 bf/cu ft (Keegan et al. 2010); green tons at 60 lb/cu ft. SDI in the summation form; maximum is the basal-area-weighted FVS Western Sierra species maximum. Sampling error at 95 % confidence with Student's t; stratum standard 40 % for tree-measurement sales (FSH 2409.12 ch. 40, 41.1).

Table 2. Stand table by 4-inch DBH class, per acre

DBH class, in	Trees	TPA	BA	cu ft	MBF
8-11.9	3	5.3	3.0	78	0.4
12-15.9	10	9.4	10.0	284	1.6
16-19.9	20	11.0	20.0	680	3.7
20-23.9	22	8.6	22.0	850	4.7
24-27.9	34	9.3	34.0	1484	8.2
28-31.9	16	3.3	16.0	711	3.9
32-35.9	34	5.4	34.0	1630	9.0
36-39.9	23	3.0	23.0	1241	6.8
40-43.9	22	2.3	22.0	1251	6.9
44-47.9	14	1.2	14.0	834	4.6
48-51.9	9	0.7	9.0	535	2.9
52-55.9	12	0.8	12.0	753	4.1
56-59.9	6	0.3	6.0	398	2.2
60-63.9	11	0.6	11.0	777	4.3
All	236	61.0	236.0	11505	63.3

Table 3. Stock table by species, per acre

Species	Trees	TPA	BA	QMD	cu ft	MBF
WF	75	17.0	75.0	28.4	3684	20.3
PP	57	15.4	57.0	26.0	2821	15.5
IC	38	11.6	38.0	24.5	1880	10.3
DF	37	10.1	37.0	25.9	1688	9.3
SP	29	6.9	29.0	27.7	1431	7.9
All	236	61.0	236.0	26.6	11505	63.3

Trees = trees tallied in the cruise; TPA, BA (sq ft/ac), cubic feet and Scribner MBF per acre by BAF expansion. Species: PP ponderosa pine, WF white fir, DF Douglas-fir, SP sugar pine, IC incense-cedar.

Unit 405: plots on the 300 ft grid (orange = QA flag)

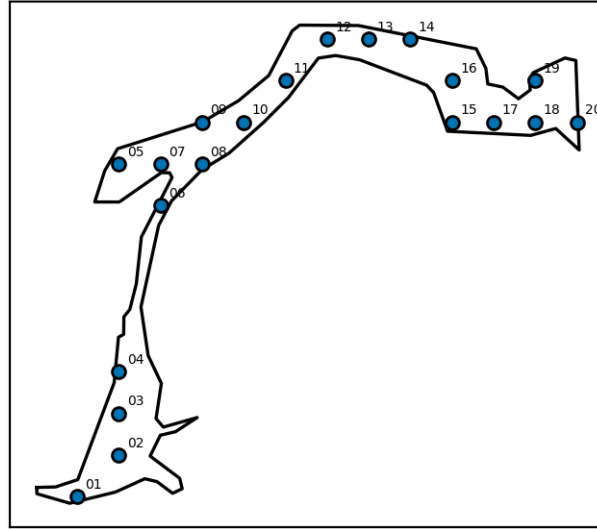


Figure 1. Plot locations with QA status.

QA findings: no issues found; unit accepted

Table 4. Plot list

Plot	Trees in	BA sq ft/ac	Slope %	Status
405-01	12	240	233	ok
405-02	12	240	267	ok
405-03	10	200	225	ok
405-04	12	240	257	ok
405-05	11	220	336	ok
405-06	11	220	203	ok
405-07	12	240	218	ok
405-08	12	240	140	ok
405-09	11	220	122	ok
405-10	12	240	239	ok
405-11	12	240	199	ok
405-12	13	260	182	ok
405-13	12	240	217	ok
405-14	15	300	247	ok
405-15	10	200	114	ok
405-16	14	280	162	ok
405-17	11	220	174	ok
405-18	11	220	174	ok
405-19	12	240	107	ok
405-20	11	220	150	ok